Chapter 10. Generative Models for Classification

Lecture 7-3 (Bayes theorem, LDA, QDA, naive Bayes, logistic regression vs LDA, comparison of classifiers)

Contents

1 Discriminant Analysis as Generative Modeling **2**
1.1 Bayes' theorem for classification . . . 2
1.2 Why discriminant analysis? . . . 2

2 Linear Discriminant Analysis **2**
2.1 One predictor . . . 2
2.2 Estimating the parameters . . . 3
2.3 Several predictors . . . 3
2.4 Adjusting the errors . . . 4

3 Quadratic Discriminant Analysis **4**

4 Naive Bayes **4**

5 Comparison of Classification Methods **5**
5.1 Logistic regression versus LDA . . . 5
5.2 QDA versus LDA, logistic regression and KNN . . . 5

6 Selected exercises **6**

1 Discriminant Analysis as Generative Modeling

Generative models describe the joint distribution

$$\Pr(X, Y) = \Pr(X \mid Y) \Pr(Y)$$

(e.g. Gaussian mixture models) and classify by Bayes' rule $\Pr(Y \mid X) = \Pr(X \mid Y) \Pr(Y) / \Pr(X)$. Discriminant analysis is the special case that assumes a specific form for $\Pr(X \mid Y)$ (e.g. Gaussian) and uses it to derive $\Pr(Y \mid X)$:

(1) model the distribution of X in each class separately, (2) flip with Bayes' theorem. Gaussian classes give

$$X \mid Y = c_k \sim N(\mu_k, \Sigma) \;\; \text{(LDA)}, \qquad X \mid Y = c_k \sim N(\mu_k, \Sigma_k) \;\; \text{(QDA)};$$

other distributions can be used as well.

1.1 Bayes' theorem for classification

Let $f_k(x)$ be the density of X in class k and $\pi_k = \Pr(Y = k)$ the prior probability. Then

$$p_k(x) = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)} \;\propto\; \pi_k f_k(x).$$

Classify to the highest posterior: $\hat{C}(x) = \arg\max_k \pi_k f_k(x)$ (the Bayes classifier of Ch. 8 with estimated ingredients). Unequal priors shift the boundary: favoring one class moves the boundary away from it.

1.2 Why discriminant analysis?

When the classes are well separated, logistic-regression estimates are surprisingly unstable (they diverge under perfect separation); LDA does not suffer from this. If n is small and X is approximately normal in each class, LDA is again more stable. LDA is popular with more than two classes and provides low-dimensional views of the data.

2 Linear Discriminant Analysis

Motivation. Logistic regression models $P(Y \mid X)$ directly. LDA instead models each class density as normal and applies Bayes' rule; it is more stable when classes are well separated or n is small.

2.1 One predictor

The Gaussian density in class k is $f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\{-\frac{1}{2}(\frac{x-\mu_k}{\sigma_k})^2\}$. LDA assumes $\sigma_1 = \cdots = \sigma_K = \sigma$. Plugging into Bayes' formula,

$$p_k(x) = \frac{\pi_k \exp\{-\frac{1}{2\sigma^2}(x-\mu_k)^2\}}{\sum_{l=1}^{K} \pi_l \exp\{-\frac{1}{2\sigma^2}(x-\mu_l)^2\}}.$$

Theorem: discriminant functions

Classifying to the largest $p_k(x)$ is equivalent to classifying to the largest

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k,$$

a *linear* function of x. With $K = 2$ and $\pi_1 = \pi_2 = 0.5$ the decision boundary is $x = (\mu_1 + \mu_2)/2$.

Proof [Supplement]. The denominator does not depend on k. Taking logs, $\log(\pi_k f_k(x)) = \log \pi_k - \frac{(x-\mu_k)^2}{2\sigma^2} + c = \log \pi_k - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + c$, and $-x^2/(2\sigma^2) + c$ is common to all classes. For two classes with equal priors, $\delta_1(x) = \delta_2(x)$ gives $x(\mu_1 - \mu_2) = (\mu_1^2 - \mu_2^2)/2$. ■

Example: simulation

$\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, $\sigma^2 = 1$: the Bayes boundary is $x = 0$, with Bayes error $\Phi(-1.5) \approx 6.7\%$. With 20 observations drawn from each class, the LDA boundary computed from the estimates lies close to 0, so its error is only slightly above the Bayes error.

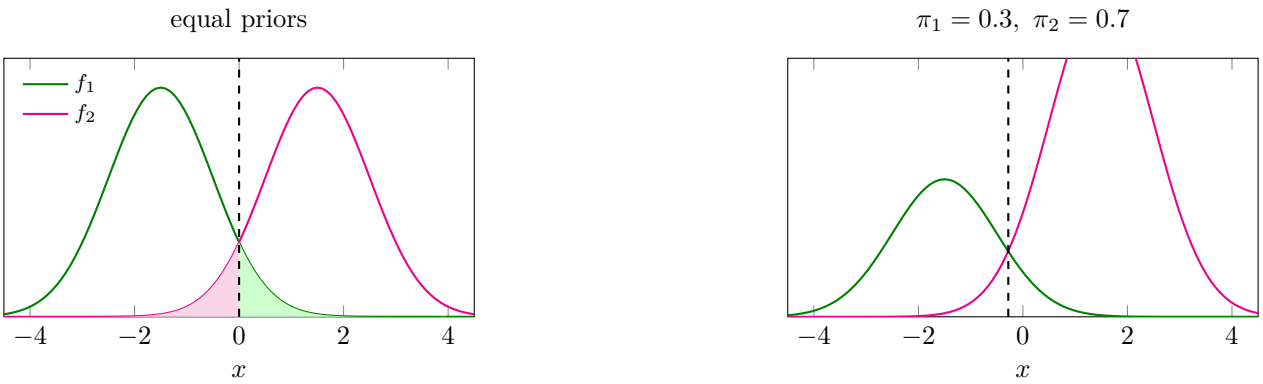


Figure 1: One-dimensional LDA with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\sigma = 1$ (curves show $\pi_k f_k$ up to a constant). Left: the Bayes boundary is $x = 0$; the shaded tails are the two misclassification probabilities. Right: favoring class 2 moves the boundary to $x = \frac{\mu_1+\mu_2}{2} + \frac{\sigma^2 \log(\pi_1/\pi_2)}{\mu_2-\mu_1} \approx -0.28$, i.e. towards class 1.

2.2 Estimating the parameters

$$\hat{\pi}_k = \frac{n_k}{n}, \qquad \hat{\mu}_k = \frac{1}{n_k}\sum_{i:y_i=k} x_i, \qquad \hat{\sigma}^2 = \frac{1}{n-K}\sum_{k=1}^{K}\sum_{i:y_i=k}(x_i - \hat{\mu}_k)^2 \quad \text{(pooled variance)},$$

plugged into $\hat{\delta}_k(x)$. These are the MLEs of the full (joint) likelihood, except for the unbiased divisor $n - K$.

2.3 Several predictors

The multivariate Gaussian density is

$$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\Big\{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big\},$$

and LDA assumes $\Sigma_k = \Sigma$ for all k.

Theorem: LDA in p dimensions

$$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k, \qquad \hat{C}(x) = \arg\max_k \delta_k(x).$$

The boundary between classes k and k' is the hyperplane

$$\Big\{x : \; x^\top \Sigma^{-1}(\mu_k - \mu_{k'}) + \log \frac{\pi_k}{\pi_{k'}} - \frac{1}{2}(\mu_k + \mu_{k'})^\top \Sigma^{-1}(\mu_k - \mu_{k'}) = 0\Big\},$$

with normal vector $\Sigma^{-1}(\mu_k - \mu_{k'})$, generally *not* in the direction of $\mu_k - \mu_{k'}$.

Proof [Supplement]. As in one dimension: $\log(\pi_k f_k(x)) = \log \pi_k - \frac{1}{2}(x-\mu_k)^\top \Sigma^{-1}(x-\mu_k) + c$, and the quadratic term $-\frac{1}{2}x^\top \Sigma^{-1} x$ is shared by all classes because Σ is common. Set $\delta_k = \delta_{k'}$. ■

With $p = 2$ and $K = 3$, the three LDA boundaries (straight lines) are close to the Bayes boundaries. **Iris data**: 4 variables, 3 species, 50 samples each; LDA classifies all but 3 of the 150 training samples correctly. **Credit (Default) data**: training error 2.75%, FPR 0.2%, FNR 75.7% (Ch. 8).

2.4 Adjusting the errors

LDA classifies to Yes if $\Pr(\text{Yes} \mid \text{balance}, \text{student}) \geq \Pr(\text{No} \mid \cdot)$, i.e. with threshold 0.5. Changing the prior of Yes, or the threshold from 0.5 to another value in $[0, 1]$, trades the two error rates. Lowering the threshold to 0.2 reduces the FNR on the Default data from 75.7% to about 41% while the overall error rises only slightly; to reduce false negatives further one may go to 0.1 or less. Logistic regression is adjusted in the same way.

3 Quadratic Discriminant Analysis

QDA allows the Σ_k to differ. The discriminant function is

$$\delta_k(x) = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^\top \Sigma_k^{-1}(x-\mu_k) + \log \pi_k = x^\top W_k x + x^\top w_k + b_k,$$

with $W_k = -\frac{1}{2}\Sigma_k^{-1}$, $w_k = \Sigma_k^{-1}\mu_k$ and $b_k = -\frac{1}{2}\mu_k^\top \Sigma_k^{-1}\mu_k - \frac{1}{2}\log|\Sigma_k| + \log \pi_k$: a quadratic form. The boundary between k and k' is the quadric

$$\{x : \; x^\top (W_k - W_{k'})x + x^\top (w_k - w_{k'}) + (b_k - b_{k'}) = 0\}.$$

Parameter count. LDA estimates Kp means and one covariance, $p(p+1)/2$ parameters; QDA estimates $Kp(p+1)/2$ covariance parameters — with $p = 50$ and $K = 3$, 3825 versus 1275. QDA works best when the class covariances differ a lot and there are enough observations to estimate them; LDA works best when the covariances are similar or data are too few (lower variance, possibly higher bias). Both perform well on many real problems. With equal covariances the Bayes boundary is linear and LDA beats QDA; with different covariances QDA is closer to the Bayes boundary.

4 Naive Bayes

Assume the inputs are independent *within each class*:

$$f(x_1, \ldots, x_p \mid y = c_k) = f(x_1 \mid c_k) f(x_2 \mid c_k) \cdots f(x_p \mid c_k).$$

Gaussian naive Bayes takes each Σ_k diagonal:

$$\delta_k(x) \;\propto\; \log\Big[\pi_k \prod_{j=1}^{p} f_{kj}(x_j)\Big] = -\frac{1}{2}\sum_{j=1}^{p}\Big[\frac{(x_j - \mu_{kj})^2}{\sigma_{kj}^2} + \log \sigma_{kj}^2\Big] + \log \pi_k.$$

Estimating a p-dimensional density is hard because the dependence among predictors must be characterized; naive Bayes eliminates it, needing only one-dimensional densities (Gaussian, kernel density, or frequency tables for categorical X_j). Although the assumption is made for convenience, it often gives decent results, especially when n is not large relative to p: the bias of a wrong independence assumption is outweighed by the large reduction in variance, and classification only needs the *ranking* of posteriors to be right, not the probabilities.

5 Comparison of Classification Methods

5.1 Logistic regression versus LDA

For LDA, the log posterior odds between classes k and K are linear:

$$\log \frac{\Pr(y = c_k \mid x)}{\Pr(y = c_K \mid x)} = x^\top \Sigma^{-1}(\mu_k - \mu_K) + \log \frac{\pi_k}{\pi_K} - \frac{1}{2}(\mu_k + \mu_K)^\top \Sigma^{-1}(\mu_k - \mu_K) = \alpha_{k0} + \alpha_k^\top x,$$

and logistic regression has linear logits by construction, $\beta_{k0} + \beta_k^\top x$. The same form — are they the same estimator? No:

- **Origin of linearity.** For LDA it is a consequence of Gaussian classes with a common covariance; for logistic regression it holds by construction.
- **Common component.** The joint density factorizes as $\Pr(X = x, Y = c_k) = \Pr(X = x) \Pr(Y = c_k \mid x)$, and for both models the second factor has the same multinomial-logit form $\exp(\beta_{k0} + \beta_k^\top x)/(1 + \sum_{k'} \exp(\beta_{k'0} + \beta_{k'}^\top x))$.
- **Different assumptions on $\Pr(X)$.** Logistic regression leaves the marginal of X arbitrary; LDA assumes the Gaussian mixture $\Pr(X = x) = \sum_k \pi_k \phi(x; \mu_k, \Sigma)$. Logistic regression makes fewer assumptions and is more general.
- **Estimation.** Logistic regression maximizes the *conditional* likelihood of Y given X, ignoring $\Pr(x)$ (fully nonparametric: the empirical distribution with mass $1/n$ at each point). LDA maximizes the *full* likelihood of $\phi(x; \mu_k, \Sigma)\pi_k$, so the marginal density does play a role.

Remarks. LDA is easier to compute. If the class densities really are Gaussian, LDA is better: logistic regression may lose about 30% efficiency asymptotically in error rate (Efron, 1975). On robustness, LDA uses all points (also those far from the boundary) to estimate Σ — more information but sensitive to outliers; logistic regression down-weights points far from the boundary (their $p_i(1 - p_i)$ weights are tiny) and is more robust.

5.2 QDA versus LDA, logistic regression and KNN

QDA is a compromise between nonparametric KNN and linear LDA/logistic regression. If the true boundary is linear, LDA and logistic regression do best; moderately nonlinear, QDA does best; more complicated, KNN (with K chosen by CV) is superior — given enough data.

Method	Boundary	Strengths / weaknesses
logistic regression	linear (or by basis expansion)	few assumptions; probabilities; unstable under separation
LDA	linear	efficient under Gaussian classes; stable with small n; outlier sensitive
QDA	quadratic	flexible covariances; many parameters
naive Bayes	depends on f_{kj}	scales to large p; independence assumption
KNN	arbitrary	nonparametric; needs much data; no interpretation

Summary

- Generative classifiers model $f_k(x)$ and π_k and classify by $\arg\max_k \pi_k f_k(x)$.
- LDA (common Σ) has linear discriminants $\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k$; QDA (class-specific Σ_k) has quadratic boundaries.
- Naive Bayes assumes conditional independence of the features, trading bias for variance.
- LDA and logistic regression share the logit form but differ in assumptions on $\Pr(X)$ and in estimation (full versus conditional likelihood).

6 Selected exercises

Motivation. ISLR Exercise 4.3 shows where the quadratic boundaries of QDA come from; a second item proves the claim that the LDA estimates are maximum-likelihood estimates; ISLR Exercise 4.7 is a small Bayes'-theorem computation.

Exercise 10.1 (QDA in one dimension; ISLR Ex. 4.3)

Let $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$ with priors π_k, $k = 1, \ldots, K$.
(a) Show that the Bayes classifier assigns x to the largest

$$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} - \log \sigma_k + \log \pi_k.$$

(b) Show that if $\sigma_k \neq \sigma_l$, then $\delta_k - \delta_l$ is a genuine quadratic, the two classes are separated by at most two points, and the class with the larger variance wins in both tails; so the Bayes classifier is not linear.

Solution.

(a) The Bayes classifier maximizes $\pi_k f_k(x)$, equivalently its logarithm

$$\log \pi_k - \tfrac{1}{2}\log(2\pi) - \log \sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}.$$

Dropping $-\frac{1}{2}\log(2\pi)$, common to all classes, and expanding the square gives δ_k. Unlike LDA, the term $-x^2/(2\sigma_k^2)$ depends on k and cannot be dropped.

(b) $\delta_k(x) - \delta_l(x) = Ax^2 + Bx + C$ with $A = \frac{1}{2\sigma_l^2} - \frac{1}{2\sigma_k^2} \neq 0$. A nonzero quadratic has at most two real roots, so the line splits into at most three intervals, alternately assigned to k and l. If $\sigma_k > \sigma_l$,

then $A > 0$ and $\delta_k - \delta_l \to +\infty$ as $|x| \to \infty$: class k wins on both tails, which no single threshold (linear rule) can produce. ■

Exercise 10.2 (LDA estimates are maximum-likelihood estimates)

Observe (x_i, y_i), $i = 1, \dots, n$, with $x_i \in \mathbb{R}$, $y_i \in \{1, \dots, K\}$, class sizes $n_k \geq 1$, and full likelihood $\prod_i \pi_{y_i} \phi(x_i; \mu_{y_i}, \sigma^2)$. Assume $W = \sum_k \sum_{i:y_i=k} (x_i - \bar{x}_k)^2 > 0$, $\bar{x}_k$ the class means.
(a) Show that the MLE is $\hat{\pi}_k = n_k/n$, $\hat{\mu}_k = \bar{x}_k$, $\hat{\sigma}^2_{\mathrm{ML}} = W/n$.
(b) Show that $\mathbb{E}W = (n - K)\sigma^2$, so the pooled $\hat{\sigma}^2 = W/(n - K)$ is the unbiased rescaling.

Solution [Supplement].

(a) The log-likelihood separates:

$$\sum_k n_k \log \pi_k \; - \; \frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_k \sum_{i:y_i=k} (x_i - \mu_k)^2.$$

Step 1 (π). For $\pi_k > 0$, $\sum_k \pi_k = 1$, Jensen's inequality for the concave log gives

$$\sum_k n_k \log \pi_k - \sum_k n_k \log \frac{n_k}{n} = n \sum_k \frac{n_k}{n} \log \frac{n\pi_k}{n_k} \leq n \log \sum_k \pi_k = 0,$$

with equality iff $n\pi_k/n_k$ is constant, i.e. $\pi_k = n_k/n$.

Step 2 (μ). For any σ^2 the last term is maximized by minimizing each $\sum_{i:y_i=k}(x_i - \mu_k)^2 = \sum_{i:y_i=k}(x_i - \bar{x}_k)^2 + n_k(\bar{x}_k - \mu_k)^2$, i.e. at $\mu_k = \bar{x}_k$; the minimum value of the double sum is W.

Step 3 (σ^2). $s \mapsto -\frac{n}{2} \log s - \frac{W}{2s}$ has derivative $(W - ns)/(2s^2)$, positive for $s < W/n$ and negative beyond, so the maximum is at W/n.

(b) Within class k the x_i are i.i.d. $N(\mu_k, \sigma^2)$, so $\mathbb{E} \sum_{i:y_i=k}(x_i - \bar{x}_k)^2 = (n_k - 1)\sigma^2$; summing over k gives $(n - K)\sigma^2$. ■

Exercise 10.3 (Predicting a dividend; ISLR Ex. 4.7)

Among companies that issued a dividend (80% of them) last year's percentage profit X has mean 10; among the others, mean 0; both have variance 36, and X is normal within each group. Find the probability that a company with $X = 4$ issues a dividend.

Solution. By Bayes' theorem with $f_k(x) \propto \exp\{-(x - \mu_k)^2/72\}$ (the common factor $1/\sqrt{72\pi}$ cancels),

$$\Pr(\text{Yes} \mid X = 4) = \frac{0.8\, e^{-(4-10)^2/72}}{0.8\, e^{-(4-10)^2/72} + 0.2\, e^{-(4-0)^2/72}} = \frac{0.8\, e^{-1/2}}{0.8\, e^{-1/2} + 0.2\, e^{-2/9}} \approx \frac{0.4852}{0.6454} \approx 0.752.$$

Although $X = 4$ is closer to the mean 0 of the non-issuers, the prior 0.8 tips the posterior towards "Yes" — the shift of the boundary by unequal priors seen in the figure above. ■